Table 7: Performance comparison of knowledge-tracing models across datasets

Verified final five-seed results · GKT pending

Dataset	Metric	DKT	DKT+	GKT	DKVMN	SAKT	LightSAKT
ASSIST2009	AUC ↑	**0.757165 ± 0.000527**	0.751202 ± 0.000876	*Pending*	0.755016 ± 0.001395	0.750322 ± 0.000648	0.751056 ± 0.000766
	F1 ↑	0.818332 ± 0.000993	0.816546 ± 0.002508	*Pending*	**0.819802 ± 0.001518**	0.817401 ± 0.000943	0.816533 ± 0.001517
	Accuracy ↑	0.736135 ± 0.001022	0.733004 ± 0.001809	*Pending*	**0.736225 ± 0.001179**	0.732718 ± 0.000669	0.732229 ± 0.000801
	MSE ↓	0.189594 ± 0.000669	0.194971 ± 0.000512	*Pending*	**0.178742 ± 0.000262**	0.180373 ± 0.000252	0.180233 ± 0.000319
	BCE ↓	0.564732 ± 0.001604	0.577168 ± 0.001301	*Pending*	**0.535401 ± 0.000548**	0.539191 ± 0.000666	0.538615 ± 0.000886
ASSIST2015	AUC ↑	**0.728235 ± 0.000327**	0.720802 ± 0.001036	*Pending*	0.727605 ± 0.000989	0.725180 ± 0.000258	0.726088 ± 0.000315
	F1 ↑	0.845662 ± 0.000528	0.842588 ± 0.000766	*Pending*	**0.847155 ± 0.000456**	0.846664 ± 0.000609	0.846282 ± 0.000392
	Accuracy ↑	0.749869 ± 0.000281	0.745974 ± 0.000619	*Pending*	**0.751034 ± 0.000445**	0.749418 ± 0.000386	0.749526 ± 0.000419
	MSE ↓	0.182124 ± 0.000394	0.187262 ± 0.000415	*Pending*	**0.170776 ± 0.000228**	0.171635 ± 0.000129	0.171233 ± 0.000106
	BCE ↓	0.547349 ± 0.001032	0.559082 ± 0.001053	*Pending*	**0.515011 ± 0.000615**	0.517255 ± 0.000332	0.516098 ± 0.000340
Algebra 2005–2006	AUC ↑	0.822181 ± 0.000683	**0.823456 ± 0.000649**	*Pending*	0.817725 ± 0.001099	0.813373 ± 0.000367	0.810590 ± 0.000422
	F1 ↑	0.872422 ± 0.000891	**0.872982 ± 0.000347**	*Pending*	0.872047 ± 0.000572	0.871012 ± 0.000372	0.870356 ± 0.000350
	Accuracy ↑	0.800524 ± 0.001175	**0.801094 ± 0.000253**	*Pending*	0.798950 ± 0.000818	0.796382 ± 0.000504	0.795574 ± 0.000639
	MSE ↓	0.149683 ± 0.000862	0.152382 ± 0.000488	*Pending*	**0.141806 ± 0.000514**	0.143309 ± 0.000206	0.144069 ± 0.000364
	BCE ↓	0.466820 ± 0.002433	0.474735 ± 0.001334	*Pending*	**0.439274 ± 0.001405**	0.443098 ± 0.000598	0.445272 ± 0.001018

Final held-out test results: mean ± sample SD over seeds 42, 123, 2026, 3407, 7777, on the same locked split.
Bold marks the best completed mean per metric: higher AUC/F1/accuracy; lower MSE/BCE. Differences are descriptive.
GKT final five-seed results are pending; no partial, primary or historical GKT scores are shown. Loss_mean is labelled BCE.